

A Practical Guide to Receiver operating characteristic

TOPIC -- RECEIVER OPERATING CHARACTERISTIC

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ROC curves beyond binary classification The extension of ROC curves for classification problems with more than two classes is cumbersome. Two common approaches for when there are multiple classes are (1) average over all pairwise AUC values and (2) compute the volume under surface (VUS). To average over all pairwise classes, one computes the AUC for each pair of classes, using only the examples from those two classes as if there were no other classes, and then averages these AUC values over all possible pairs. When there are c classes there will be $c(c - 1) / 2$ possible pairs of classes. The volume under surface approach has one plot a hypersurface rather than a curve and then measure the hypervolume under that hypersurface. Every possible decision rule that one might use for a classifier for c classes can be described in terms of its true positive rates ($TPR_1, \dots, TPR_c$). It is this set of rates that defines a point, and the set of all possible decision rules yields a cloud of points that define the hypersurface. With this definition, the VUS is the probability that the classifier will be able to correctly label all c examples when it is given a set that has one randomly selected example from each class. The implementation of a classifier that knows that its input set consists of one example from each class might first compute a goodness-of-fit score for each of the c^2 possible pairings of an example to a class, and then employ the Hungarian algorithm to maximize the sum of the c selected scores over all $c!$ possible ways to assign exactly one example to each class. Given the success of ROC curves for the assessment of classification models, the extension of ROC curves for other supervised tasks has also been investigated. Notable proposals for regression problems are the so-called regression error characteristic (REC) Curves and the Regression ROC (RROC) curves. In the latter, RROC curves become extremely similar to ROC curves for classification, with the notions of asymmetry, dominance and convex hull. Also, the area under RROC curves is proportional to the error variance of the regression model.

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History The ROC curve was first developed by electrical engineers and radar engineers during World War II for detecting enemy objects in battlefields, starting in 1941, which led to its name ("receiver operating characteristic"). It was soon introduced to psychology to account for the perceptual detection of stimuli. ROC

analysis has been used in medicine, radiology, biometrics, forecasting of natural hazards, meteorology, model performance assessment, and other areas for many decades and is increasingly used in machine learning and data mining research.

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An alternative to the ROC curve is the detection error tradeoff (DET) graph, which plots the false negative rate (missed detections) vs. the false positive rate (false alarms) on non-linearly transformed x- and y-axes. The transformation function is the quantile function of the normal distribution, i.e., the inverse of the cumulative normal distribution. It is, in fact, the same transformation as zROC, below, except that the complement of the hit rate, the miss rate or false negative rate, is used. This alternative spends more graph area on the region of interest. Most of the ROC area is of little interest; one primarily cares about the region tight against the y-axis and the top left corner – which, because of using miss rate instead of its complement, the hit rate, is the lower left corner in a DET plot. Furthermore, DET graphs have the useful property of linearity and a linear threshold behavior for normal distributions. The DET plot is used extensively in the automatic speaker recognition community, where the name DET was first used. The analysis of the ROC performance in graphs with this warping of the axes was used by psychologists in perception studies halfway through the 20th century, where this was dubbed "double probability paper".

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These figures are the TOC and ROC curves using the same data and thresholds. Consider the point that corresponds to a threshold of 74. The TOC curve shows the number of hits, which is 3, and hence the number of misses, which is 7. Additionally, the TOC curve shows that the number of false alarms is 4 and the number of correct rejections is 16. At any given point in the ROC curve, it is possible to glean values for the ratios of false alarms $\frac{\text{false alarms}}{\text{false alarms} + \text{correct rejections}}$ and hits $\frac{\text{hits}}{\text{hits} + \text{misses}}$. For example, at threshold 74, it is evident that the x coordinate is 0.2 and the y coordinate is 0.3. However, these two values are insufficient to construct all entries of the underlying two-by-two contingency table.

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Several studies criticize certain applications of the ROC curve and its area under the curve as measurements for assessing binary classifications when they do

not capture the information relevant to the application. The main criticism to the ROC curve described in these studies regards the incorporation of areas with low sensitivity and low specificity (both lower than 0.5) for the calculation of the total area under the curve (AUC), as described in the plot on the right. According to the authors of these studies, that portion of area under the curve (with low sensitivity and low specificity) regards confusion matrices where binary predictions obtain bad results, and therefore should not be included for the assessment of the overall performance. Moreover, that portion of AUC indicates a space with high or low confusion matrix threshold which is rarely of interest for scientists performing a binary classification in any field. Another criticism to the ROC and its area under the curve is that they say nothing about precision and negative predictive value. A high ROC AUC, such as 0.9 for example, might correspond to low values of precision and negative predictive value, such as 0.2 and 0.1 in the $[0, 1]$ range. If one performed a binary classification, obtained an ROC AUC of 0.9 and decided to focus only on this metric, they might overoptimistically believe their binary test was excellent. However, if this person took a look at the values of precision and negative predictive value, they might discover their values are low. The ROC AUC summarizes sensitivity and specificity, but does not inform regarding precision and negative predictive value.